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Trigonometry

An Original Olympiad Practice Set

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Original content. These 8 problems were written fresh for this set — they are **not** copied from Titu Andreescu's books or any competition. Eight original olympiad-style trigonometry problems spanning conditional identities, telescoping trig series, the bounded-amplitude principle for max/min, and the deep symmetry of regular-7-gon angles. Each is built so the decisive move is a structural insight — a clever pairing, a telescope, a Chebyshev/root-of-unity argument, or a convexity bound — rather than brute computation. Every closed-form answer was checked symbolically/numerically. Free to share for study.

Problems

1. DIFFICULTY 2/5 · PRODUCT-TO-SUM, SYMMETRIC ANGLE PAIRING, POWER-REDUCTION

Evaluate

$$S = \sin^2 10^\circ + \sin^2 50^\circ + \sin^2 70^\circ.$$

2. DIFFICULTY 2/5 · TRIPLE-ANGLE IDENTITY, FACTORING, PRODUCT OF TANGENTS

Without a calculator, prove that

$$\tan 20^\circ \ \tan 40^\circ \ \tan 80^\circ = \tan 60^\circ,$$

and state the numerical value.

3. DIFFICULTY 2/5 · AMPLITUDE-PHASE REDUCTION, BOUNDING VIA SINGLE SINUSOID, RANGE OF A FUNCTION

Find the exact set of values taken by

$$f(x) = 12\sin x - 5\cos x + 4, \qquad x \in \mathbb{R},$$

and the values of x where the maximum is attained.

4. DIFFICULTY 3/5 · TELESCOPING, COTANGENT DIFFERENCE IDENTITY, DOUBLE-ANGLE

Let θ be such that none of $\theta, 2\theta, \ldots, 2^n\theta$ is a multiple of π. Find a closed form for

$$T_n = \sum_{k=1}^{n} \frac{1}{\sin(2^k\theta)} = \frac{1}{\sin 2\theta} + \frac{1}{\sin 4\theta} + \cdots + \frac{1}{\sin 2^n\theta}.$$

5. DIFFICULTY 3/5 · TELESCOPING PRODUCT, SINE DOUBLE-ANGLE, LIMIT

For $0 < \theta < \pi$ prove the product formula

$$\prod_{k=1}^{n} \cos\frac{\theta}{2^k} = \frac{\sin\theta}{2^n \sin(\theta/2^n)},$$

and deduce the value of the infinite product $\displaystyle\prod_{k=1}^{\infty} \cos\frac{\theta}{2^k}$.

6. DIFFICULTY 4/5 · ROOTS OF UNITY, REGULAR 7-GON SYMMETRY, PRODUCT-TO-SUM TELESCOPING

Prove that

$$\cos\frac{\pi}{7} - \cos\frac{2\pi}{7} + \cos\frac{3\pi}{7} = \frac{1}{2}.$$

7. DIFFICULTY 4/5 · MINIMAL POLYNOMIAL OF TAN(KΠ/7), VIETA'S FORMULAS, IMAGINARY PART OF (COS+I SIN)^7

Show that

$$\tan^2\frac{\pi}{7}+\tan^2\frac{2\pi}{7}+\tan^2\frac{3\pi}{7}=21.$$

8. DIFFICULTY 5/5 · CONSTRAINED OPTIMIZATION, TRIANGLE COTANGENT IDENTITY, SOS INEQUALITY, SYMMETRY

Let A, B, C be the angles of a triangle, so $A+B+C=\pi$ and $A, B, C\in(0,\pi)$. Determine the minimum value of

$$g=\cot^2 A+\cot^2 B+\cot^2 C,$$

and prove it is the minimum.

Solutions

1. Answer: \dfrac{3}{2}

Use power reduction $\sin^2\theta = \frac{1-\cos 2\theta}{2}$:

$$S = \frac{3}{2} - \frac{1}{2}\big(\cos 20^\circ + \cos 100^\circ + \cos 140^\circ\big).$$

The insight is to pair the last two by sum-to-product about their mean 120°: $\cos 100^\circ + \cos 140^\circ = 2\cos 120^\circ \cos 20^\circ = -\cos 20^\circ$. Hence

$$\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = \cos 20^\circ - \cos 20^\circ = 0,$$

and therefore $S = \dfrac{3}{2}$.

Second method. Complex-exponential view: the doubled angles can be arranged so that $\mathrm{Re}\big(e^{i\,20^\circ} + e^{i\,100^\circ} + e^{i\,140^\circ}\big)$ is computed via $\cos 100^\circ + \cos 140^\circ = -\cos 20^\circ$, giving 0; thus the cosine sum vanishes and $S = \frac{3}{2}$.

2. Answer: \sqrt{3}

The hidden engine is the factorization $\tan 3\theta = \tan\theta\,\tan(60^\circ - \theta)\,\tan(60^\circ + \theta)$. With $\theta = 20^\circ$, $60^\circ - \theta = 40^\circ$ and $60^\circ + \theta = 80^\circ$, so $\tan 60^\circ = \tan 20^\circ \tan 40^\circ \tan 80^\circ$. To justify the identity, write $\tan 3\theta = \tan\theta \cdot \dfrac{3 - \tan^2\theta}{1 - 3\tan^2\theta}$. With $t = \tan\theta$, the factorizations $3 - t^2 = (\sqrt{3} - t)(\sqrt{3} + t)$ and $1 - 3t^2 = (1 - \sqrt{3}t)(1 + \sqrt{3}t)$, together with $\tan(60^\circ \mp \theta) = \dfrac{\sqrt{3} \mp t}{1 \pm \sqrt{3}t}$, give $\tan(60^\circ - \theta)\tan(60^\circ + \theta) = \dfrac{3 - t^2}{1 - 3t^2} = \dfrac{\tan 3\theta}{\tan\theta}$. Multiplying by $\tan\theta$ proves the claim. Since $\tan 60^\circ = \sqrt{3}$, the product is $\sqrt{3}$.

Second method. Direct evaluation uses $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}$ and $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$; their ratio is the desired product, equal to $\sqrt{3}$.

3. Answer: [-9,\,17]

Any $a\sin x + b\cos x$ is a single sinusoid of amplitude $\sqrt{a^2+b^2}$: there is a phase φ with $a\sin x + b\cos x = \sqrt{a^2+b^2}\sin(x+\varphi)$. Here $a = 12, b = -5$, amplitude $\sqrt{144+25} = 13$, so $f(x) = 13\sin(x+\varphi) + 4$ with $\tan\varphi = -\frac{5}{12}$. Since $\sin(x+\varphi) \in [-1,1]$, f ranges over $[4-13, 4+13] = [-9, 17]$. The maximum 17 occurs when $\sin(x+\varphi) = 1$; at that point $(\sin x, \cos x) = \big(\frac{12}{13}, -\frac{5}{13}\big)$, i.e. $x = \pi - \arctan\frac{12}{5}$ (second quadrant), plus integer multiples of 2π.

Second method. Calculus: $f'(x) = 12\cos x + 5\sin x = 0 \Rightarrow \tan x = -\frac{12}{5}$; the two critical directions give $(\sin x, \cos x) = \pm\big(\frac{12}{13}, -\frac{5}{13}\big)$, hence $f = 4 \pm 13 = \{17, -9\}$, and by continuity the range is the whole interval $[-9, 17]$.

4. Answer: T_n=\cot\theta-\cot\!\left(2^{n}\theta\right)

The decisive observation is that a cosecant of a doubled angle is a difference of cotangents:

$$\cot\alpha-\cot 2\alpha=\frac{\cos\alpha}{\sin\alpha}-\frac{\cos 2\alpha}{\sin 2\alpha}=\frac{2\cos^2\alpha-\cos 2\alpha}{\sin 2\alpha}=\frac{2\cos^2\alpha-(2\cos^2\alpha-1)}{\sin 2\alpha}=\frac{1}{\sin 2\alpha}.$$

Setting $\alpha=2^{k-1}\theta$ gives $\dfrac{1}{\sin 2^k\theta}=\cot 2^{k-1}\theta-\cot 2^k\theta$. Summing for $k=1,\dots,n$ telescopes:

$$T_n=\sum_{k=1}^{n}\left(\cot 2^{k-1}\theta-\cot 2^k\theta\right)=\cot\theta-\cot 2^n\theta.$$

Second method. Induction: base $n=1$ is $\csc 2\theta=\cot\theta-\cot 2\theta$. Step: $T_{n+1}=(\cot\theta-\cot 2^n\theta)+(\cot 2^n\theta-\cot 2^{n+1}\theta)=\cot\theta-\cot 2^{n+1}\theta$.

5. Answer: \displaystyle\prod_{k=1}^{\infty}\cos\frac{\theta}{2^{k}}=\frac{\sin\theta}{\theta}

Multiply $P_n=\prod_{k=1}^{n}\cos(\theta/2^k)$ by the smallest sine $\sin(\theta/2^n)$ and cascade $\sin 2x=2\sin x\cos x$:

$$P_n\sin\frac{\theta}{2^n}=\cos\frac{\theta}{2}\cdots\cos\frac{\theta}{2^{n-1}}\cdot\tfrac{1}{2}\sin\frac{\theta}{2^{n-1}}.$$

The new sine merges with the preceding cosine, and the collapse repeats n times to give $P_n\sin(\theta/2^n)=\frac{1}{2^n}\sin\theta$, so

$$P_n=\frac{\sin\theta}{2^n\sin(\theta/2^n)}.$$

For the infinite product, $2^n\sin(\theta/2^n)=\theta\cdot\dfrac{\sin(\theta/2^n)}{\theta/2^n}\to\theta$, hence $\prod_{k=1}^{\infty}\cos\frac{\theta}{2^k}=\dfrac{\sin\theta}{\theta}$. (At $\theta=\pi/2$ this is Viète's formula $\frac{2}{\pi}=\cos\frac{\pi}{4}\cos\frac{\pi}{8}\cdots$.)

Second method. Induction: assuming $P_{n-1}=\dfrac{\sin\theta}{2^{n-1}\sin(\theta/2^{n-1})}$, substitute $\sin(\theta/2^{n-1})=2\sin(\theta/2^n)\cos(\theta/2^n)$ into $P_n=P_{n-1}\cos(\theta/2^n)$ to cancel the cosine and obtain $P_n=\dfrac{\sin\theta}{2^n\sin(\theta/2^n)}$.

6. Answer: \dfrac{1}{2}

First reduce to a positive-sign sum. Since $\cos\frac{5\pi}{7}=-\cos\frac{2\pi}{7}$, the target equals $C:=\cos\frac{\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{5\pi}{7}$. Now use the seven solutions of $z^7=-1$, namely $z=e^{i(2k+1)\pi/7}$, $k=0,\dots,6$, which sum to 0 (coefficient of z^6 in z^7+1 is 0). Their real parts pair as conjugates, and $k=3$ gives $\cos\pi=-1$:

$$2\left(\cos\tfrac{\pi}{7}+\cos\tfrac{3\pi}{7}+\cos\tfrac{5\pi}{7}\right)+(-1)=0\implies C=\tfrac{1}{2}.$$

Thus $\cos\frac{\pi}{7}-\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}=C=\frac{1}{2}$.

Second method. Multiply C by $2\sin\frac{\pi}{7}$ and use $2\sin\frac{\pi}{7}\cos\frac{(2j-1)\pi}{7}=\sin\frac{2j\pi}{7}-\sin\frac{(2j-2)\pi}{7}$; the sum telescopes to $\sin\frac{6\pi}{7}=\sin\frac{\pi}{7}$, so $C=\frac{1}{2}$. Then $\cos\frac{5\pi}{7}=-\cos\frac{2\pi}{7}$ gives the stated form.

7. Answer: 21

Build a polynomial whose roots are exactly the three quantities. Expanding $(\cos\theta + i\sin\theta)^7$ and forming $\tan 7\theta = \dfrac{\mathrm{Im}}{\mathrm{Re}}$ gives, with $t = \tan\theta$,

$$\tan 7\theta = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6}.$$

For $\theta = \frac{k\pi}{7}$ we have $\tan 7\theta = 0$, so $t = \tan\frac{k\pi}{7}$ satisfies the numerator: $t\big(t^6 - 21t^4 + 35t^2 - 7\big) = 0$. The nonzero roots $\tan\frac{k\pi}{7}$ $(k = 1, \ldots, 6)$ occur in $\pm$ pairs since $\tan\frac{(7-k)\pi}{7} = -\tan\frac{k\pi}{7}$; hence $u = t^2$ takes the three values $\tan^2\frac{\pi}{7}, \tan^2\frac{2\pi}{7}, \tan^2\frac{3\pi}{7}$, the roots of

$$u^3 - 21u^2 + 35u - 7 = 0.$$

By Vieta the sum of roots is the negation of the u^2-coefficient, namely 21. Hence the sum of squares is 21. (Bonus: $\sum\sec^2\frac{k\pi}{7} = 21 + 3 = 24$.)

Second method. Derive the same cubic in $u = \tan^2$ from $\cos 7\theta = 0$ at $\theta = \frac{(2k-1)\pi}{14}$ (cotangent version) and read the leading coefficients; Vieta again yields sum $= 21$.

8. Answer: 1

At the equilateral triangle $A = B = C = \frac{\pi}{3}$, $\cot\frac{\pi}{3} = \frac{1}{\sqrt{3}}$, so $g = 3 \cdot \frac{1}{3} = 1$. We prove $g \geq 1$. The key tool is the triangle identity

$$\cot A\cot B + \cot B\cot C + \cot C\cot A = 1,$$

which follows from $\cot C = -\cot(A + B) = \dfrac{1 - \cot A\cot B}{\cot A + \cot B}$ after clearing denominators. Combine it with the SOS inequality $x^2 + y^2 + z^2 \geq xy + yz + zx$ (equivalently $\frac{1}{2}\sum(x - y)^2 \geq 0$) at $x = \cot A, y = \cot B, z = \cot C$:

$$\cot^2 A + \cot^2 B + \cot^2 C \geq \cot A\cot B + \cot B\cot C + \cot C\cot A = 1.$$

Equality needs $\cot A = \cot B = \cot C$, i.e. $A = B = C = \frac{\pi}{3}$, which is admissible. Hence the minimum value is 1. (The expression is unbounded above as any angle $\to 0$, so only a minimum exists.)

Second method. Lagrange multipliers on $\sum\cot^2 A$ subject to $\sum A = \pi$ give $\cot A\csc^2 A = \cot B\csc^2 B = \cot C\csc^2 C$; the unique interior solution is $A = B = C = \frac{\pi}{3}$, yielding $g = 1$, confirmed as the minimum by the boundary blow-up.